

A Coarse-Grained Model Describing the Critical Micelle Concentration of Perfluoroalkyl Surfactants in Ionic Aqueous Phase

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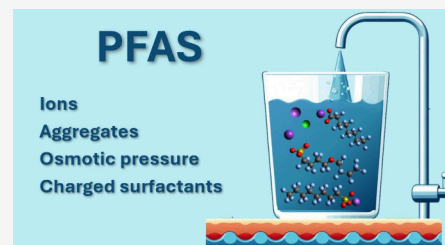


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Supporting Information

ABSTRACT: In this study, dissipative particle dynamics (DPD) simulations were employed to determine the critical micelle concentration (CMC) of perfluoroalkyl and polyfluoroalkyl substances (PFAS) in ionic aqueous solutions. This approach provides precise CMC data for PFAS surfactants in the presence of various ionic species, thereby addressing a gap in the current literature. Additionally, this study contributes to the development of open-source molecular force fields for charged perfluorinated compounds, which are currently limited. These models incorporate hydration free energy values obtained from density functional theory (DFT) and account for ionic interactions through a well-established linear relationship. Hydrophobic interactions between the surfactant tail and water were fine-tuned to match the CMC of chosen surfactants. Then, the DPD models successfully predicted CMC values for a diverse range of surfactants, including those based on hydrocarbons and PFAS, demonstrating the ability to represent realistic salinities encountered in natural waters. Experimental validation of the methodology was conducted using sodium *n*-nonyl sulfate (SNS) and sodium *n*-dodecyl sulfate (SDS) via interfacial tension measurements, confirming the accurate representation of the CMC changes with salinity. This study enhances our understanding of the behavior of PFAS surfactants in ionic aqueous solutions and provides a valuable tool for predicting CMC values in complex environmental systems.



INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) are employed in a wide array of applications, including textiles, packaging, household products, cosmetics, personal care products, paints, pesticides, and firefighting foams.¹ PFAS pose environmental concerns due to their persistence, bioaccumulation, and harmful effects on human health and ecosystems. The initial focus of the authorities was put on long-chain PFAS ($\geq 8C$), including perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS).^{1,2} Recently, the carcinogenic potential of PFOA and PFOS has been established by the International Agency for Research on Cancer (IARC).³ PFOA is now classified as carcinogenic to humans (Group 1), while PFOS is classified as possibly carcinogenic to humans (Group 2B).³ To circumvent regulatory restrictions on long chains, short-chain PFAS ($\leq 7C$) are employed as substitutes, attracting attention as their prevalence in the environment increases.¹ Adsorption is the primary method for PFAS removal, but studies often overlook the influence of organic matter and salinity in natural waters.² However, salt composition and ionic strength demonstrably affect the interfacial adsorption of ionic surfactants.⁴ To predict the behavior of PFAS more accurately in these environments, the development of molecular models that incorporate salinity and charged interactions is essential.⁵ Interestingly, a subset of PFAS exhibits unique surfactant properties. For instance, perfluoroalkyl carboxylic acid (PFCA)

and perfluorosulfonic acid (PFSA) are PFAS with a molecular structure comprising both a hydrophobic and lipophobic fluoroalkyl chain and a hydrophilic functional group.¹ Accurate models of such ionic surfactants could also help to develop new alternatives to remediate oil-contaminated soil and groundwater.⁶ In addition, these models could predict partition coefficients, which play a crucial role in the fate and transport in underground vadose and saturated zones.^{7,8}

When introduced into an appropriate medium at concentrations exceeding their critical micelle concentration (CMC), surfactants display a remarkable propensity to self-assemble into a variety of structures, including spherical micelles, bilayers, vesicles, and lamellar layers.⁹ Using chemoinformatics tools, the CMC of surfactants can be predicted with a quantitative structure–activity relationship (QSPR).^{10,11} Alternatively, Moriarty et al.¹² developed graph neural networks (GNNs) which effectively predict CMCs of surfactants. At the mesoscopic scale, the simulation of surfactant aggregates represented by coarse-grained (CG) beads of atoms and

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using dissipative particle dynamics (DPD) has been the subject of extensive investigations.^{13–34} This coarse-grained method is well-suited for describing the formation of aggregates that occurs within the microsecond time scale, which is beyond the reach of atomistic molecular dynamics (MD) simulations. In fact, the self-assembly process is primarily influenced by the equilibrium between the entropy loss from surfactants and counterions and the energy gain from the tails, while electrostatic repulsion has a lower impact.³⁵ All of these factors can be incorporated into a DPD simulation. DPD models can be employed in their uncharged state to represent nonionic surfactants or simplified ionic surfactants, incorporating electrostatic interactions implicitly in the repulsive interactions.^{14,16,26,29,36} Otherwise, DPD models for ionic surfactants can be further refined by incorporating charged beads and electrostatic interactions.^{13,15,17–19,21,23,24,27,28,33} No prior research, to the best of our knowledge, has investigated the self-aggregation of PFAS surfactants using DPD simulations. This lacuna may be explained by two key factors: the difficulty of finding experimental data to fit with and the early development of incorporating electrostatics into meso-scale DPD simulations. However, DPD studies have successfully simulated the behavior of nonaggregated PFAS in aqueous environments. For instance, Feng et al.³⁷ carried out simulations on PFAS to study their diffusion and mobility from water to soil organic matter with an uncharged model.³⁷ We hypothesize that incorporating electrostatic interactions into the model and calibrating the bead parameters using experimental CMC data will result in a more accurate DPD representation of PFAS. Electrostatic interactions in DPD simulations are typically modeled using a smeared charge approximation, where the charge is distributed around the bead center rather than being concentrated at a single point.^{38,39}

Parameterizing the repulsive coefficients in charged DPD simulations necessitates rigorous methods to ensure an accurate representation of ionic interactions. In the absence of a universally accepted approach, ions are often represented as charged water beads.^{19,23,27,28} This simplifies ion interactions but neglects their complexity.³⁶ Consequently, researchers have proposed the use of advanced parametrization methods. Mayoral et al.⁴⁰ adjust repulsive coefficients based on Flory–Huggins χ parameters, while Vishnyakov et al.³⁰ and others^{41,42} use infinite dilution activity coefficients. Recently, Nieto-Draghi et al.⁴³ proposed a parametrization procedure for electrolytes in an aqueous solution based on osmotic pressure and hydration free energy.⁴³ Alternatively, Lavagnini et al.³⁶ utilizes Setschenow coefficients to adjust interactions between salt ions and surfactants, but this is limited to nonionic surfactants and a specific DPD model with variable cutoffs.

Experimentally, CMC is usually determined via surface tension or conductivity.^{44,45} In DPD simulations, the CMC can be identified by monitoring the free surfactant concentration through cluster analysis.^{29,36} Cluster analysis offers the advantage of revealing aggregate size distribution, similar to light scattering.^{29,36,46} An alternative methodology is to employ thermodynamic properties, such as osmotic pressure or chemical potential, which undergo a change of slope with concentration upon micelle formation and allows for determining the CMC.^{47,48}

This study aims to integrate the methodologies proposed by Lavagnini et al.³⁶ and Nieto-Draghi et al.⁴³ to establish a comprehensive parametrization of charged beads in DPD

simulations. We first used alkyl-based surfactant experimental data for strategy development and then applied the methodology to reproduce available CMC experimental data on PFAS molecules. In practice, repulsive interactions involving ions (ionic–ionic and ionic–water) are derived from solvation Gibbs free energies. The validation is achieved by parametrizing and predicting CMCs of established alkyl sulfate surfactants by adjusting the repulsion between surfactants' tail beads and water. Osmotic pressure and cluster analysis are employed to determine the CMC of sodium *n*-nonyl sulfate (SNS). The validity of the SNS model is further assessed by evaluating the effect of salinity on CMC compared to our experimental data. Finally, parametrization and CMC prediction are performed for PFAS.

MATERIALS AND METHODS

Experimental CMC Determination. The CMC of sodium *n*-nonyl sulfate (SNS), and sodium *n*-dodecyl sulfate (SDS) was determined experimentally by using the Wilhelmy plate method. SDS ($\text{C}_{12}\text{H}_{25}\text{NaO}_4\text{S}$, CAS: 151-21-3, MW: 288.38 g/mol, Sigma-Aldrich, 99%) and SNS ($\text{C}_9\text{H}_{19}\text{NaO}_4\text{S}$, CAS: 1072-15-7, MW: 246.3 g/mol, Alfa Aesar 99%) were used as received. The tensiometer used was a Kruss K100 equipped with a platinum plate with a resolution of 0.01 mN/m. Measurements were performed at room temperature of 25 °C. The measurement was automatic, with one point taken every minute until stabilization over 5 consecutive points or 1 h of measurement. The uncertainty associated with the concentration of the samples prepared by dilution was estimated to be 0.72%. NaCl crystals were pyrolyzed at 300 °C for 2 days to remove any impurities. The CMC was determined at the intersection of the linear regression of the surface tension plateau and the linear regression of the linear regime preceding the plateau (see Figure S7-2, Supporting Information), with an error estimated at 3.4%.

DPD Principles and Simulation Details. Dissipative particle dynamics (DPD) is a coarse-grained simulation method that models the behavior of complex fluids using soft potentials.^{20,49} This allows DPD simulations to be performed on large and complex systems such as surfactant solutions. The equation of motion in DPD is decomposed in $m_i \frac{d\vec{v}_i}{dt} = \vec{F}_{ij}^C + \vec{F}_{ij}^D + \vec{F}_{ij}^R + \vec{F}_{ij}^S + \vec{F}_{ij}^{\text{el}}$, where m_i is the mass of particle i , $\vec{v}_i$ is its velocity, and $\vec{F}_{ij}^C$, $\vec{F}_{ij}^D$, $\vec{F}_{ij}^R$, $\vec{F}_{ij}^S$, and $\vec{F}_{ij}^{\text{el}}$ are the conservative, dissipative, random, spring, and Coulombic forces acting on the particle, respectively. The spring forces are only added to connect beads and form molecules. A more detailed account of the forces involved, and their decomposition, is provided in Section S1.1 of the Supporting Information.

DPD simulations use “smeared charges” (Slater distribution) to model electrostatics. This distributes charge smoothly over beads, avoiding issues with point charges like infinite potential and unrealistic ion pairs.^{36,38,43,50,51} We employed the formulation of González-Melchor et al.³⁸ for the smeared electrostatic force. All of the simulation parameters and the conversion table to real units are available in Section S1.1 of the Supporting Information. DPD simulations were carried out using the LAMMPS software,⁵² and visualizations were done using the OVITO software.⁵³

A series of DPD models were developed for each surfactant family, with specific repulsive coefficients and coarse-grained representations. The surfactant chains are represented as simplified DPD beads connected through harmonic bonds with optimized rigidity.²⁹ This approach captures the chain's influence on micelle formation and CMC.^{20,29}

The surfactant structures were mapped to DPD beads with volumes corresponding to approximately three water molecules. The bead types reflected hydrophobicity and charge, and their volumes were determined using the Materials Studio QSAR Models module⁵⁴ (see Table S2-1).

Parameterization of the Repulsive Coefficients. We initially calculated all pairwise repulsive parameters, as detailed in Table 1.

Table 1. Summary of the Methodology Used to Set the Repulsive Parameters between Beads for Charged Surfactants in Ionic Solutions^a

repulsive parameter	origin	equation
water/water	water compressibility	$a_{ww} = \frac{k_B T}{2\alpha\rho} (\kappa^{-1} N_m - 1)$
like/like	same as water	$a_{ii} = a_{ww}$
tail/water	fitted with the CMC using osmotic pressure	-
charged/water	DFT	$a_{C-W} = 0.132720\Delta G_{hyd} + 123.148$
cationic/anionic	DFT	$a_{ij} = c_1\Delta\Delta G_{hyd} + c_0$
charged/nonionic	DFT	$\Delta a_{C-N} = a_{C-N} - a_{ww} = b\chi \approx b\Delta E/RT$

^a ΔE is the interaction energy between two beads approximated by DFT calculations. b , c_0 , and c_1 are constant numbers. ΔG_{hyd} is the solvation free energy of a charged bead. $\Delta\Delta G_{hyd}$ is the difference of solvation free energy between two charged beads. χ is the Flory–Huggins interaction parameter.

Subsequently, we optimized the repulsion between the tail beads and the water beads, as this interaction plays a crucial role in driving hydrophobicity and significantly affects the CMC of the surfactant.^{12,29,36,55} Sections S3 and S4 in the Supporting Information provide comprehensive lists of all the parameters used. The parameters provided in this work are surfactant independent, and they are consistent for all molecules sharing the same functional group. For instance, any molecule containing an alkyl group with three carbons (a C3 bead) or a carboxylic group (RCOO^-) will use the same set of interaction parameters in the system. Except for the tail–water interaction, the repulsive parameters for the uncharged–uncharged beads and like beads were set equal to the water–water repulsion. Although DFT calculations of these interactions could have been performed to refine the model, their overall contribution to the CMC was considered to be negligible; therefore, the default value for water was used. In contrast, charged beads exhibit an additional interaction force that requires adjustment of their repulsive coefficients with water and other charged beads. To address this, we employed a linear relationship (eq 1) proposed by Nieto-Draghi et al.⁴³ between repulsive parameters (a_{ij}) and the solvation free energy (ΔG_{hyd} , $\text{kJ}\cdot\text{mol}^{-1}$) of ionic species.

$$a_{\text{ion-water}} = 0.132720\Delta G_{hyd} + 123.148 \quad (1)$$

In their study, Nieto-Draghi et al.⁴³ employed measured individual ion activities⁵⁶ to construct eq 1. Subsequently, these activities were converted into partial osmotic pressures in order to parametrize ion–water interactions. However, as recently emphasized by Warren,⁵⁷ the precise thermodynamic meaning of osmotic pressure in a system is only derived from the combined contributions of both counterions (the cation and the anion) across the membrane (refer to the methodological description below).⁵⁷ Nevertheless, the optimization procedure for eq 1 requires the input of initial guess values for both anions and cations. In the absence of alternative information for initializing the optimization of ion–water interactions, Nieto-Draghi et al.⁴³ proposed that partial osmotic pressures could be used as initial values. This assumption yielded reliable repulsion parameters, as demonstrated below with the variation of CMC with added salinity. Consequently, the aforementioned approach was retained again in our parametrization strategy.

This relationship enabled us to determine the repulsive parameters for the charged beads and water beads. For oppositely charged beads, we used another linear relationship with the difference of solvation free energy ($\Delta\Delta G_{hyd}$), as described by Nieto-Draghi et al.⁴³ and given in eq 2:

$$a_{ij} = c_1\Delta\Delta G_{hyd} + c_0 \quad (2)$$

with c_0 and c_1 being constants of the linear regression for a given cation interacting with anions.

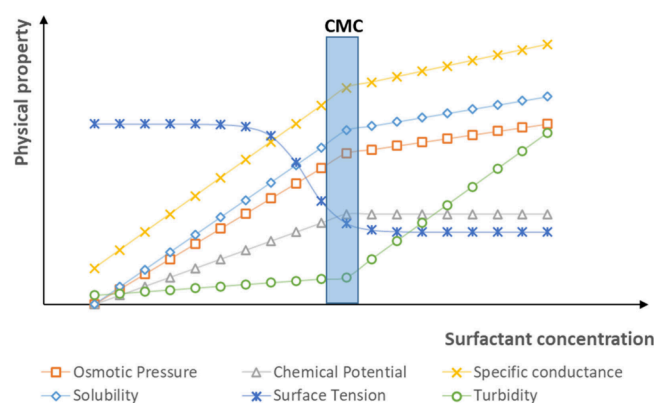
The repulsive parameters for charged–uncharged beads were derived from DFT calculations, further described in Section S3.3 of the Supporting Information.

DPD Methods for CMC Determination. Surfactant molecules self-assemble above the CMC, significantly altering the physical

properties of the solution (Figure 1). Thermodynamically, the CMC manifests as a distinct change in the slope of the osmotic pressure or chemical potential with increasing surfactant concentration. These properties exhibit concentration profiles with a sharp break at the CMC.⁵⁸ Additionally, the phase model of micelles allows for CMC determination from the change in surfactant chemical potential.⁵⁹ Monte Carlo methods permit the calculation of the chemical potential in DPD simulations. However, this method presents large fluctuations due to the soft potentials inherent in DPD and the presence of charged species. This approach is therefore unsuitable for the purposes of this study.

Cluster analyses of concentrated surfactant solutions can also be performed using DPD simulations.^{17–19} Above the CMC, the free surfactant concentration is expected to remain constant at the CMC value. However, DPD simulations present a challenge in estimating the free surfactant concentration due to crowding effects. The phenomenon of crowding is observed when the total surfactant concentration is increased beyond the CMC, resulting in a decrease in the concentration of free surfactant.^{60–65} This behavior is unrealistic and thus requires a mathematical correction.²⁴

Osmotic Pressure Method Using Virtual Walls. Mimicking osmotic pressure experiments in simulations (DPD or atomistic) involves virtual walls interacting with solutes.^{43,66,67} These walls confine solutes by using a recalling harmonic potential. We specifically implemented a new LAMMPS fix to create this recalling force called “wall/harmonic/returned” and a new pair style to accelerate the charged DPD calculations on GPU named “dpd/coul/slater/long”.⁶⁸ The recalling force eliminates ambiguity regarding the control volume and failures on particle–wall overlap in comparison with the already implemented repulsive force. Our approach allows for a soft force while maintaining a defined control volume, independent of the harmonic force strength. Depending on the statistical ensemble, the corresponding framework of the obtained osmotic pressures may vary. As mentioned by Nieto-Draghi et al.,⁴³ the statistical ensemble

**Figure 1.** Examples of ideal CMC breakpoints by using different physical properties.

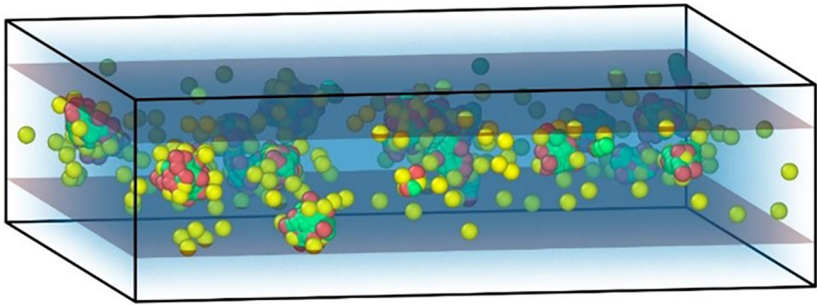


Figure 2. Snapshot of the simulation cell for PFOS (H_1T_4) using our model PALSNa. A high concentration of 24 mM is used to verify aggregates do not interact with both walls at the same time. Here, the cell dimensions are $78.30 \times 78.30 \times 23.00$. Color map of beads: H: red (SO_3^-), T: green ($\text{CF}_2\text{--CF}_2$), NA: yellow (Na^+ ; $3\text{H}_2\text{O}$), W: transparent blue ($3\text{H}_2\text{O}$). The harmonic walls are represented as transparent red planes.

Table 2. Comparative Analysis of Experimental CMC Values with DPD Simulations (This Work) and QSPR Predictions from Ref 11

chemical formula and abbreviation	CG representation (parameters)	experimental CMC (mM) min–max	simulated CMC (mM)	QSPR CMC prediction (mM)
Alkyl Sulfate Surfactants				
$\text{C}_{12}\text{H}_{25}\text{OSO}_3\text{Na}$ SDS	H_1T_4 (ALS)	$8.0^{d,e} \text{--} 8.2^{9,29}$	$7.2^f \pm 0.8$	7.2 ± 1.1
$\text{C}_9\text{H}_{19}\text{OSO}_3\text{Na}$ SNS	H_1T_3 (ALS)	$60\text{--}69^{d29,72,76}$	66 ± 4	78.4 ± 11.8
$\text{C}_6\text{H}_{13}\text{OSO}_3\text{Na}$ SHS	H_1T_2 (ALS)	$290\text{--}520^{29,72,76}$	$240^a \pm 22$	648.1 ± 97.2
Perfluoroalkyl Sulfonate Surfactants (PFSA)				
$\text{C}_8\text{F}_{17}\text{SO}_3\text{Na}$ Na-PFOS	H_1T_4 (PALSNa)	$1.07\text{--}8.00^{75,78,80\text{--}83}$ 1.8^{e71}	$1.8^f \pm 0.4$	1.8 ± 0.1
$\text{C}_8\text{F}_{17}\text{SO}_3\text{K}$ K-PFOS	H_1T_4 (PALSK)	$0.89\text{--}1.32^{80,83,86}$	1.3 ± 0.5	0.9 ± 0.1
$\text{C}_6\text{F}_{13}\text{SO}_3\text{K}$ K-PFHxS	H_1T_3 (PALSK)	12^{b81}	8 ± 1	4.9 ± 0.7
$\text{C}_4\text{F}_9\text{SO}_3\text{K}$ K-PFBS	H_1T_2 (PALSK)	22^{b81}	$24^a \pm 3$	22.5 ± 3.4
Perfluoroalkyl-Carboxylate Surfactants (PFCA)				
$\text{C}_9\text{F}_{19}\text{COONa}$ Na-PFDA	H_1T_4 (PALC2)	$0.85^b \text{--} 0.91^{c45,78,79,82}$	1.7 ± 0.2	2.1 ± 0.3
$\text{C}_8\text{F}_{17}\text{COONa}$ Na-PFNA	H_1T_4 (PALC1)	$10.66\text{--}12.15^{45,78,79,82}$	3.5 ± 0.7	7.2 ± 1.1
$\text{C}_7\text{F}_{15}\text{COONa}$ Na-PFOA	H_1T_3 (PALC2)	$29\text{--}36^{71,79,85,86}$	14 ± 1.4	24.1 ± 3.6
$\text{C}_6\text{F}_{13}\text{COONa}$ Na-PFHpA	H_1T_3 (PALC1)	$82\text{--}87^{45,78,79,82,88}$	$39^a \pm 8$	74.9 ± 11.2
$\text{C}_9\text{F}_{19}\text{COOH}$ H-PFDA	H_1T_4 (PALC2)	$0.3\text{--}1.4^{78,82,84}$	1.4 ± 0.1	0.9 ± 0.1
$\text{C}_8\text{F}_{17}\text{COOH}$ H-PFNA	H_1T_4 (PALC1)	$1.3\text{--}8.5^{75,82,84}$	1.7 ± 0.1	3.0 ± 0.4
$\text{C}_7\text{F}_{15}\text{COOH}$ H-PFOA	H_1T_3 (PALC2)	$4\text{--}10^{71,78,82,84}$	10.9 ± 0.9	9.5 ± 1.4
$\text{C}_6\text{F}_{13}\text{COOH}$ H-PFHpA	H_1T_3 (PALC1)	$10\text{--}30^{78,82,84}$	22 ± 6.6	29.3 ± 4.4
Perfluoroalkyl Ether Sulfonate Surfactants (PFAES)				
$\text{C}_8\text{F}_{17}\text{SO}_4\text{K}$ K-F404	HTOTT (PAES)	1.8^{e83}	$1.8^f \pm 0.4$	1.8 ± 0.3
$\text{C}_6\text{F}_{17}\text{SO}_4\text{K}$ K-PES	HOT (PAES)	-	$37^a \pm 3$	18.6 ± 2.8

^aSecond breakpoint used. ^bLimited precision. ^cInsolubility concerns at 25 °C. ^dOur experimental CMCs. ^eCMC used as fitting target. ^fFitted value.

determines the osmotic pressure framework. Isothermal–isobaric ensembles correspond to Lewis–Randall conditions, while the canonical ensemble requires McMillan–Mayer scaling, which is

negligible at low ionic concentrations and is thus ignored here. The osmotic pressure (Π_i) of the solute i was determined using the mean

force necessary to maintain the solute in the control volume following eq 3:

$$\Pi_i = \frac{\langle F_i \rangle}{\text{area}} \quad (3)$$

where $\langle F_i \rangle$ is the time averaged force exerted by the walls on the solute i and “area” is the total membrane surface.

The osmotic coefficient (ϕ_i) of the solute i was also calculated using eq 4 in DPD units:

$$\phi_i = \frac{\Pi_i}{\Pi_i^{\text{ideal}}}; \quad \Pi_i^{\text{ideal}} = \frac{N_{\text{solute}}}{V_c} \quad (4)$$

where N_{solute} is the number of solute molecules in the control volume V_c .

Instead of using cubic cells, we constructed cobblestone shape simulation boxes, with one example given in Figure 2. This shape improves the statistical reliability of the measurements in diluted conditions. As illustrated in Figure 2, high concentrations of surfactants were tested using this shape to verify that large aggregates are not in contact with both walls at the same time and that the osmotic pressure is the same as that with a cubic volume control. The construction procedure is available in Section S1.2 of the Supporting Information. To ensure the reliability of the uncertainty estimates, three independent simulations were employed for the optimization runs.

A sample of input and post-treatment scripts are freely available at <https://ifpen-gitlab.appcollaboratif.fr/ifpen-ms-external/dpd-cmc-osp>.

Cluster Analyses. The simulations of SNS at different salting conditions were analyzed using the UMMAP post-treatment tool.²² These simulations use cubic boxes in the isothermal–isobaric ensemble with a side length of 44 in DPD units. Both the equilibration and production phases were conducted for 1.5 million steps. The UMMAP tool has two main parameters, the submicellar cutoff (pm) and the cluster analysis cutoff (r_{ac}). Submicelles are aggregates of surfactants that are smaller than micelles. To adjust this cutoff, we followed the methodology of Anderson et al.²⁴ and others.^{24,29,69} All of the detailed steps are available in Section S7.3.

CMC Breakpoint Determination Using Osmotic Curves. Building on prior work by Santos et al.,⁵⁸ we employed osmotic pressure versus concentration curves to determine the CMC. While traditional methods rely on surfactant volume fraction (not readily accessible in DPD simulations), our approach leverages concentration, a more suitable descriptor. It is acknowledged that the evolution of osmotic pressure above CMC can be nonlinear, as evidenced in the existing literature.^{47,48} However, the van't Hoff law ensures linearity at low concentrations.^{47,48,58}

To identify the CMC breakpoint, we utilized a Bayesian piecewise linear model with optimal breakpoints via the “piecewise-regression” package.⁷⁰ Ideally, there is agreement between the resulting CMC from the osmotic pressure curve and the CMC from the osmotic coefficient curve. This involves an initial linearity followed by a deviation (see Figure 1). However, the reported CMCs are derived only from the osmotic pressure curves for better precision (see the explanations provided in the discussion of K-PFBS).

RESULTS AND DISCUSSION

Literature Review of Experimental CMCs. A comprehensive literature review was conducted to extract experimental CMC values for PFAS and alkyl sulfates. Note that in a very recent work, we investigated the development of QSPR models generated with a support vector machine algorithm and a group counts representation of surfactants including 60 PFAS.¹¹ These models will be used for comparison purposes hereafter. In this study, the collected CMC values^{9,29,45,71–84} served either as fitting targets for the DPD models or as comparative/validation values and are presented in Table 2.

Notably, several review papers failed to distinguish between the counterion form of the surfactant used during CMC measurements, whether it was a salt or a free acid. This distinction is crucial, as the counterion significantly impacts the CMC. For instance, we can find CMCs of PFOA in pure water that range from 8 to 9 mM,^{71,78,80,82} while the CMC of Na-PFOA salt is reported in the range of 29 to 36 mM.^{71,79,85,86} The observed increase in the CMC of PFOA upon the inclusion of a sodium counterion can be attributed to the screening of charges at the aggregate surface, thereby demonstrating the impact of ionic interactions on the aggregation process. Furthermore, the increase in concentration of counterions can also lower the repulsive forces between ionic surfactant headgroups, which will decrease the CMC.⁸² Additionally, temperature information was often omitted, despite the fact that perfluoro-*n*-alkanoates (PFCA) with more than 8 carbons exhibit a Krafft point above room temperature.^{45,71,79} This indicates that these surfactants require a temperature above room temperature to form micelles; otherwise, they should precipitate with concentration. To achieve greater precision, a more rigorous approach would involve strict pH control and refining the true concentration using filtration followed by high-performance liquid chromatography (HPLC) measurement.⁸⁷ Please bear in mind the diversity of experimental conditions and reported precisions, which can significantly compromise the reliability of the comparative data employed in this study.^{9,29,45,71–74,74–84}

Modeling Alkyl Sulfate Surfactant CMCs. The alkyl sulfate surfactant model (ALS) was initially calibrated against sodium *n*-dodecyl sulfate (SDS) to accurately reproduce the experimental CMC. After refining the repulsive coefficient, the simulated CMC matched the experimental value within the confidence interval (Table 2), validating the model's accuracy. Nevertheless, a key limitation of the osmotic pressure technique is its inability to provide information regarding the aggregation number. Consistent with Mai et al.,²⁹ we employed the optimized value of $a_{H-T} = 104 k_B T$ for the ALS model. We concur with their findings that this repulsive parameter yields the correct aggregation number ($N_{ag} = 33$) without impacting the CMC for SNS solutions analyzed using the UMMAP tool.²²

The model's quality is further corroborated by its capacity to reproduce the observed trend of the experimental osmotic coefficient as a function of SDS concentration (see Figure 3). Our simulated osmotic coefficient has the benefit of reaching low concentrations, usually inaccessible through experiments, and showing a first linear regime. However, the error is larger at low concentrations, as expected, due to the reduced collision frequency of surfactant molecules with the walls. At concentrations above the CMC, our simulated osmotic coefficient exceeds the available experimental data, suggesting potential model limitations in capturing complex interactions in concentrated systems, presumably dominated by the formation of vesicles.

To further evaluate the accuracy and transferability of the model, we predicted the CMC of SNS and SHS. The corresponding osmotic curves are shown in Section S6.1 of the Supporting Information. The SNS osmotic pressure presents two linear regimes as a function of its concentration, and its osmotic coefficient presents only one linear regime below the CMC. Both curves present a CMC breakpoint at 66 ± 4 mM, which is in great concordance with our experimental result of 69 mM.

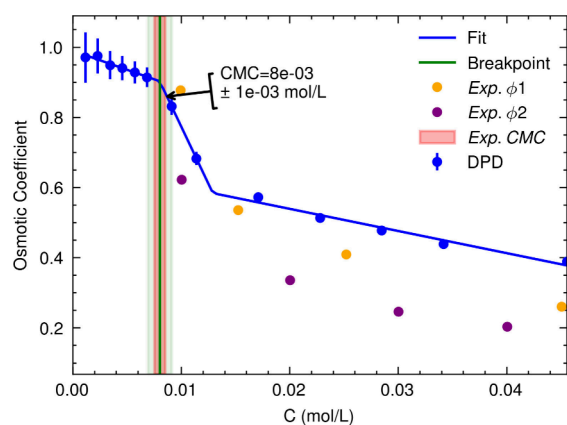


Figure 3. Osmotic coefficient evolution with the concentration of SDS using the ALS model. The experimental CMC range is depicted by a transparent red region and is derived from refs 9 and 29. Our predicted CMC range is indicated by the green shaded region. The experimental osmotic coefficients are taken from refs 89 and 90.

Determining the CMC of SHS is challenging due to its atypical osmotic pressure profile (see Figure S6-5). Unlike other surfactants, SHS exhibits two distinct linear regimes in the osmotic pressure curve, suggesting different aggregation behaviors. The second breakpoint, identified through the alignment of the osmotic pressure curve with the osmotic coefficient curve, is proposed as the authentic CMC. Further corroboration is provided by cluster analysis, which indicates the formation of submicelles below this point. For additional details, please refer to Section S6.1.3 in the Supporting Information. In light of this case, it is recommended that both the pressure and coefficient curves be analyzed in order to ensure the most reliable determination of the CMC.

The atypical behavior of SHS, likely due to its shorter chain length, may necessitate model refinement with additional parameters or a smaller coarse-grained level for improved agreement with the experimental CMC values.

Our piecewise linear regression analysis of osmotic curves effectively identifies transition points in surfactant behavior, even if they do not directly reflect specific configurations.

Salinity Effect on the CMC. This research aims to develop DPD models for predicting the CMC of ionic surfactants (alkyl and PFAS) in realistic salinities. Experimental CMCs of alkyl sulfates (SNS, SDS) at varying NaCl concentrations (Figure S7-1) are used to validate the methodology. Surface tension curves (e.g., Figure S7-2) were used to determine CMCs via the Wilhelmy plate method. The obtained CMCs exhibit a logarithmic trend with the NaCl concentration.

To validate our parametrization, we employed the ALS model, without modification for added salt, to predict the CMC of SNS solutions with varying NaCl concentrations (0, 0.1, 0.2, and 0.5 M). Consistent with experimental findings (Figure 4), our DPD model accurately predicts the decrease in SNS CMC with increasing NaCl concentration. This close agreement between methods validates our parametrization approach, thereby indicating accurate representation of ionic interactions. The salting-out effect arises from added ions mitigating electrostatic repulsion between headgroups in micellized surfactants.^{12,91} The CMC–NaCl concentration curve follows the relationship of eq 5 for ionic surfactants:⁹¹

$$\ln(C_{\text{CMC}}) = \ln(C_{\text{CMC}_0}) - A \ln(1 + BC_{\text{ion}}) \quad (5)$$

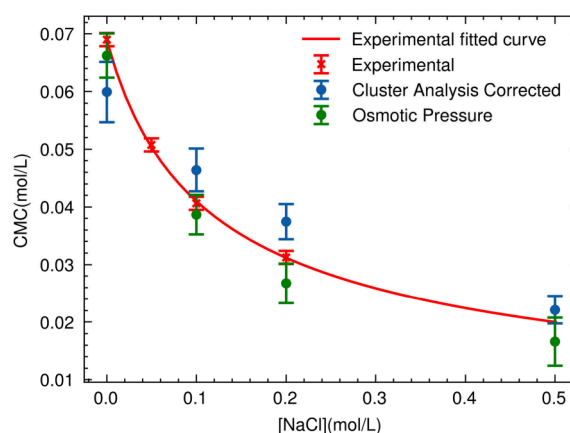


Figure 4. Evolution of the SNS CMC with the salting effect of NaCl. Comparison of two methods of CMC determination from DPD simulations with our experimental points. Line corresponds to the fitting of experimental data to eq 5.

where CMC_0 is the CMC at a salt concentration (C_{ion}) of zero and A and B are two fitting parameters.

Beyond predicting the CMCs of pure surfactants, the presented models can also be applied to surfactant mixtures in ionic solutions, which are notoriously difficult to predict.¹⁹ The models can be combined in solution mixtures with the sole prerequisite being the calculation of the repulsive parameters between the charged functions of different surfactant families (e.g., sulfonic acid with carboxylic acid beads).

The present study compared two methods for determining the CMC: osmotic pressure and cluster analysis. Osmotic pressure emerged as the most accessible approach, yielding accurate CMC values with minimal parameter adjustments. This advantage arises from the elimination of the need for tuning and corrections in the cluster analysis. Moreover, osmotic coefficients have been demonstrated to be a valuable tool in determining the CMC in specific scenarios and are experimentally accessible values. However, the method necessitates the allocation of additional computing resources. For example, the CMC determination of SHS with 0.1 M NaCl running on Intel(R) Xeon(R) Gold 6140 CPU @ 2.30 GHz required 7776 CPU hours for the cluster analysis method (3 concentrations with 3 replications each). The osmotic pressure method required three times more resources, with 23 328 CPU hours (27 concentrations, 3 runs each).

Perfluoroalkyl Sulfonate Surfactants. We employed the same parametrization strategy for PFSA surfactants. Optimization of the water/tail parameter for the PALSNa model yielded an optimal repulsion of 99.905 in DPD units (details in Section S4.2). Osmotic pressure simulations resulted in a CMC of Na-PFOS at 1.8 ± 0.4 mM, consistent with experimental values (1.07 to 8 mM).^{75,78,80–83} The concentration range encompassed values from 0.3 to 37 mM, a range that extends well beyond the CMC (see Figure S6-9). With regard to SDS (Figure 3), it is anticipated that the osmotic coefficient of Na-PFOS will be underestimated for concentrations significantly exceeding its CMC. However, the unavailability of experimental osmotic coefficients renders any conclusion challenging. In order to ensure that concentrated conditions accurately reproduce the type of organization of the aggregates, the structure factor could be used for direct comparison with SANS experiments.⁹² Indeed, concentrated PFAS in solutions can aggregate in rod-like

structures and lamellar sheets; however, this phenomenon is out of the scope of the present work.^{92,93} As demonstrated by Gong et al.,⁹⁴ the chain stiffness of surfactants plays a crucial role in the formation of vesicles and could be optimized accordingly.

The optimized parameters of PALSNa are reused in the PALSK model to enable the full prediction of other CMCs containing potassium counterions. The initial prediction concerns the CMC of K-PFOS, which is in close proximity to that of Na-PFOS, exhibiting a slight reduction from 1.8 to 1.3 mM, respectively.^{80,83} The PALSK model predicted a higher CMC (8 mM) for K-PFHxS (shorter chain) due to the reduced hydrophobicity. This value differs from the literature value (12 mM), likely due to limitations in data precision.^{81,95}

Validation is prevented by a lack of additional literature values. Still, the value from the QSPR model (4.9 mM) supports the DPD result. Finally, the PALSK model predicted the CMC for K-PFBS. This short-chain surfactant presents an osmotic curve with a linear regime at low concentrations, followed by a region of curvature and then another linear regime at high concentrations (see Figure S6-14). The first breakpoint in the osmotic pressure curve gives a satisfactory result of CMC at 24 ± 3 mM, close to the literature value of 22 mM⁸¹ and the QSPR prediction of 22.3 mM. However, the osmotic coefficient curve (Figure S6-15) exhibits a breakpoint at 90 ± 10 mM, which is considered to be excessively high. Upon further analysis of the osmotic pressure curve, a subtle inflection is observed above the identified breakpoint. Due to the nature of the osmotic coefficient construction, its numerical precision is decreased, and this example demonstrates that it is less sensitive to subtle changes in the curve.

Sodium alkyl sulfonate surfactants present a logarithmic relationship between the CMC and the chain length (N_c),⁹⁶ given in eq 6. This empirical link was originally found by Kleven.⁹⁷

$$\log(\text{CMC}) = A - BN_c \quad (6)$$

where A is a constant depending on the nature of the surfactant's head and B is another constant usually equal to or close to $\log(2)$. This logarithmic relationship is explained by the Gibbs free energy of micellization, which is mainly impacted by the transfer free energy.^{86,98,99} Examination of the limited experimental data for K-PFSA CMCs presented in Figure 5 also suggests a logarithmic trend of the CMC with chain length. The fitted parameters of eq 6 are given in our paper about the QSPR model.¹¹ The RMSD of ± 2.58 mM between the experimental and simulated CMCs indicates a good agreement with the observed decrease in CMC for perfluoroalkyl sulfonates.

Perfluoroalkyl-Carboxylate Surfactants. Two models, PALC1 and PALC2, were employed to simulate perfluoroalkyl-carboxylate surfactants (PFCA). The parameters were derived from DFT calculations with the exception of the optimized repulsive parameter between the tail and water, which was derived from the PALSK model. PALC1, designed for even-carbon PFCA (Na-PFNA, Na-PFHpA), possesses a head bead (H) corresponding to COO^- but deviates from volume homogeneity assumptions due to its 60% water bead volume. PALC2, for odd-carbon PFCA (Na-PFDA, Na-PFOA), incorporates a CF_2 group in the head bead and achieves better volume homogeneity.

DPD simulations employing the PALC1 and PALC2 models systematically underestimated the CMC of various PFCA in

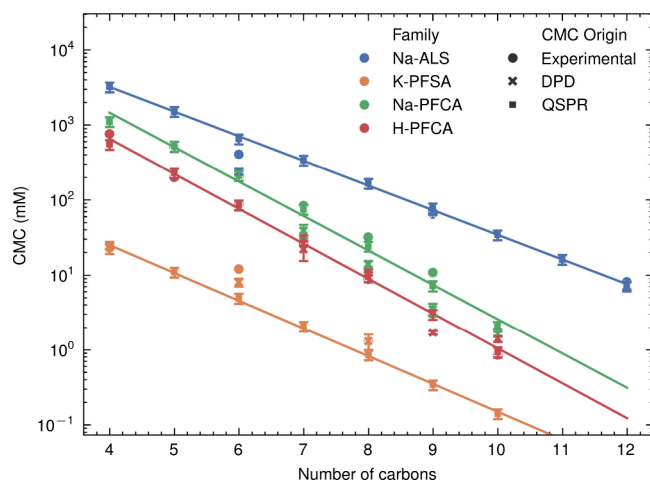


Figure 5. Chain length dependence of CMC values for Na-PFCA, H-PFCA, Na-ALS, and K-PFSA surfactant families. Experimental data (dots) are presented alongside the DPD simulation results (arrows) and QSPR predictions (connected cube ticks). Lines represent linear regression of eq 6 with the QSPR predictions.

comparison with experimental data (see Table 2). This discrepancy was observed for both H_1T_4 and H_1T_3 molecules, although their pressure curves exhibited distinct regimes with clear CMC breakpoints (see Section S6.3). Notably, the QSPR model also deviated from experimental values, particularly for Na-PFDA (10 carbons). These discrepancies suggest potential inaccuracies in the Na-PFCA experimental data.

In order to identify the cause of underestimated CMCs of Na-PFCA, a systematic adjustment of repulsive parameters (excluding water tail) was conducted. It was observed that weakening water–head repulsion led to an increase in CMCs (improved solubility), while strengthening head–counterion repulsion had the same effect (reduced micelle stabilization). Head–tail repulsion was found to affect the aggregation number but not CMC. Head–head interactions were found to be dominated by Coulombic forces, rendering repulsion irrelevant. Tail–tail interactions impact the average aggregation number (N_{ag}) of surfactants. It should be noted that the default repulsion of water–water applied to tail–tail beads results in $N_{ag} = 18.15 \pm 1.58$, which is in excellent agreement with the average value reported in the literature ($N_{ag} = 17.5$).¹⁰⁰ Despite all of these adjustments, the simulated CMCs remained below the reported experimental values.

To gain further insights, the well-characterized family of H-PFCA was examined. Here, the QSPR model demonstrated a superior fit to experimental CMCs compared to Na-PFCA (see Figure 5). Employing the same DPD models (PALC1 and PALC2) with a charged water bead counterion to represent H-PFCA yielded predictions that closely matched both the experimental and QSPR values. This successful application to H-PFCA validates the reliability of the optimization procedure and the extrapolation of parameters to other surfactant families. However, the observed discrepancies regarding Na-PFCA may be attributed to the pH of the solutions, which was not accounted for in this study and not reported in the referenced experimental studies.

Perfluoroalkyl Ether Sulfonate Surfactants. The methodology's versatility was further demonstrated by optimizing the CMC of perfluoroalkyl ether sulfonate (PFES) surfactants. The repulsive parameter between the

ether bead and water bead was optimized using F404, a well-characterized PFES. This resulted in a predicted CMC that closely matched the experimental value (Table 2). The method was then applied to commercially available K-PES (potassium perfluoro(2-ethoxyethane)sulfonate). Due to the short carbon chain, a clear CMC breakpoint was absent in the pressure curve (see Figure S6-34). However, the predicted CMC aligned with the expected trend of decreasing CMCs with longer chains and offered a valuable estimate in the absence of a definitive experimental value.

Our CMC determination method can also be extended to branched surfactants and unusual chemicals, such as the perfluoro surfactants studied here. The QSPR models are highly effective at providing tendencies and are valuable tools with which to fit the DPD models with. In cases involving mixtures of surfactants and salts with fewer experimental data for comparison, our DPD models are arguably the optimal choice.

CONCLUSION

This work presents a systematic methodology for parametrizing dissipative particle dynamics (DPD) simulations of charged surfactants in ionic solutions. The proposed methodology extends the parametrization of ion beads by Nieto-Draghi et al.⁴³ to the parametrization of surfactant charged beads. The remaining repulsive coefficients of hydrophobic surfactant tails with water are fitted until reproduction of the critical micelle concentration (CMC). A thermodynamic approach is employed, whereby the osmotic pressure evolution with concentration allows for determining the CMC. The osmotic pressure is obtained by using the virtual walls method in DPD simulations at fixed total surfactant concentrations. In comparison to the conventional cluster analysis methodology, our approach is less susceptible to parametrization and mathematical corrections, thereby facilitating a more robust and reliable determination of CMC values. Despite its advantages, the osmotic pressure technique requires a higher computational cost compared to cluster analyses and may not be suitable for short-chain surfactants due to the gradual evolution of osmotic pressure with surfactant concentration. In such cases, the osmotic coefficient can be employed as an additional tool to identify the most probable location of the CMC point.

In conclusion, our models effectively predict the CMCs of several perfluorinated surfactants, which are occasionally absent from the literature or lacking precision regarding the surrounding ionic species. Moreover, the models accurately capture the behavior of charged surfactants in ionic solutions. We successfully reproduced the salting effect on surfactant CMCs and demonstrated that our technique is sensitive to the counterion type. With these models and the established methodology in place, we can envisage predicting other environmental properties, including partition coefficients and contaminant adsorption, with a precise consideration of the realistic salinity found in natural waters. In order to gain further insight into the effects of bending and harmonic constants on short surfactants, it would be beneficial to conduct further research in this area. This is because short surfactants may exhibit distinct stiffness and degrees of freedom compared to long surfactants.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.4c04293>.

Conversion table; bead volumes; DFT calculation results; model parameter sets and mappings; cell shape algorithm and contact probability; osmotic pressure and osmotic coefficient curves as a function of the concentration for all the surfactants studied; experimental and simulated salting effect over the SNS CMC; and osmotic pressure and cluster aggregation (PDF)

Raw_results: osmotic pressure and coefficient data as a function of the concentration for all the surfactants studied (XLSX)

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E.B. performed the simulations. The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

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ABBREVIATIONS

ALS, alkyl sulfate surfactants; DPD, dissipative particle dynamics; DFT, density functional theory; CG, coarse-grained; CMC, critical micelle concentration; F404, 1,1,2,2,3,3,4,4-octafluoro-4-(perfluorobutoxy)butane-1-sulfonate; SNS, sodium *n*-nonyl sulfate; SDS, sodium *n*-dodecyl sulfate; SHS, sodium *n*-hexadecyl sulfate; PES, perfluoro(2-ethoxyethane)-sulfonate; PFOS, perfluorooctanesulfonate; PFDA, perfluorododecanoate; PFNA, perfluorononanoate; PFHxS, perfluorohexanesulfonate; PFBS, perfluorobutanesulfonate; PFOA, perfluorooctanoate; PFHpA, perfluoroheptanoate; PFSA,

perfluoroalkyl sulfonate surfactants; PFCA, perfluoroalkyl-carboxylate surfactants; PFAES, perfluoroalkyl ether sulfonates; QSPR, quantitative structure–activity relationship; RMSD, root mean square deviation

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